WORKED SOLUTION

STEP II 2020, Question 9

Mechanics • Colliding Projectiles

STEP II 2020, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

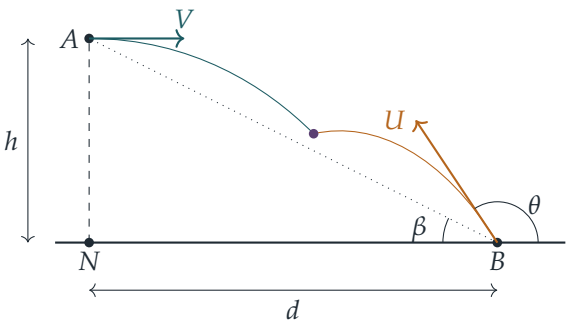
www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Put both particles in one coordinate system and write down where each is at time t. Both feel the same downward acceleration g, so the $-\frac{1}{2}gt^2$ terms cancel when we compare heights: relative to each other, the particles move in straight lines at constant velocity. That makes the collision time very simple, and every later part is a short consequence of it.

Solution

Model both particles as point masses moving freely under gravity, with no air resistance. Take N as origin, the x-axis along NB (so $B = (d, 0)$, $A = (0, h)$) and the y-axis vertically upwards. The particle from A moves towards B and the particle from B moves towards N, so their horizontal velocities are opposite; this is the set-up the given result describes.



At time t after projection, the particle from A is at

$$x_A = Vt, \qquad y_A = h - \tfrac{1}{2}gt^2,$$

and the particle from B is at

$$x_B = d - Ut\cos\theta, \qquad y_B = Ut\sin\theta - \tfrac{1}{2}gt^2.$$

They collide at time T if both coordinates agree. The heights agree when

$$h - \tfrac{1}{2}gT^2 = UT\sin\theta - \tfrac{1}{2}gT^2 \quad \Longrightarrow \quad T = \frac{h}{U\sin\theta},$$

which is the only possible collision time ($\sin\theta > 0$ as θ is acute). The horizontal positions must agree at this same time:

$$VT = d - UT\cos\theta \quad \Longrightarrow \quad (V + U\cos\theta)\,\frac{h}{U\sin\theta} = d.$$

Multiplying by $U\sin\theta$ gives $Vh + Uh\cos\theta = Ud\sin\theta$, and dividing by U,

$$d\sin\theta - h\cos\theta = \frac{Vh}{U},$$

as required.

Part (i)

From the right-angled triangle ANB, $\tan\beta = h/d$, with $0 < \beta < \frac{1}{2}\pi$. Since V, h, U are positive, the result just shown gives $d\sin\theta - h\cos\theta > 0$. Dividing by $d\cos\theta > 0$ (as θ is acute),

$$\tan\theta > \frac{h}{d} = \tan\beta.$$

Both angles lie in $(0, \frac{1}{2}\pi)$, where tan is strictly increasing, so $\theta > \beta$.

Part (ii)

> **Ask yourself** Nothing in the algebra so far has used the ground. What physical condition have we not yet imposed?

The collision must happen before either particle reaches the ground, so the common height at time T must be at least 0:

$$h - \tfrac{1}{2}gT^2 \geq 0 \quad \Longleftrightarrow \quad h \geq \frac{gh^2}{2U^2\sin^2\theta} \quad \Longleftrightarrow \quad U^2\sin^2\theta \geq \frac{gh}{2}.$$

(We divided by $h > 0$ and multiplied by $U^2\sin^2\theta > 0$.) Since $U\sin\theta > 0$, taking positive square roots gives

$$U\sin\theta \geq \sqrt{\frac{gh}{2}}.$$

Equality corresponds to the particles meeting just as they reach the ground.

Part (iii)

Write $r = AB = \sqrt{d^2 + h^2}$, so that $d = r\cos\beta$ and $h = r\sin\beta$. Then

$$d\sin\theta - h\cos\theta = r(\sin\theta\cos\beta - \cos\theta\sin\beta) = r\sin(\theta - \beta),$$

and the collision condition becomes

$$r\sin(\theta-\beta)=\frac{Vr\sin\beta}{U} \quad\Longrightarrow\quad \frac{U}{V}=\frac{\sin\beta}{\sin(\theta-\beta)}.$$

By part (i), $\theta-\beta>0$; also $\theta-\beta<\theta<\frac{1}{2}\pi$. So $0<\sin(\theta-\beta)<1$, and since $\sin\beta>0$,

$$\frac{U}{V}=\frac{\sin\beta}{\sin(\theta-\beta)}>\sin\beta.$$

The final part

The height of the collision is $h-\frac{1}{2}gT^2$, so

$$\text{collision above } \tfrac{1}{2}h \iff h-\tfrac{1}{2}gT^2>\tfrac{1}{2}h \iff gT^2<h.$$

The particle from B has vertical velocity $U\sin\theta-gt$, so

$$\text{it is rising at time } T \iff U\sin\theta-gT>0 \iff gT<U\sin\theta.$$

Ask yourself How can we compare $gT<U\sin\theta$ with $gT^2<h$?

Use the collision time itself: $T=h/(U\sin\theta)$ means $U\sin\theta=h/T$. So, multiplying by $T>0$,

$$gT<U\sin\theta \iff gT<\frac{h}{T} \iff gT^2<h.$$

Both conditions are equivalent to $gT^2<h$, so the particles collide above height $\frac{1}{2}h$ if and only if the particle from B is still moving upwards at the collision.

Checks

- Take $g=9.8$, $h=10$, $d=20$, $U=20$, $\theta=\frac{1}{4}\pi$. The collision condition gives $V=U(d\sin\theta-h\cos\theta)/h=10\sqrt{2}\approx 14.14$, and $T=1/\sqrt{2}\approx 0.707$ s. Then both particles are at $x=10$, $y=10-4.9\times\frac{1}{2}=7.55$, above $\frac{1}{2}h=5$; the particle from B has vertical velocity $10\sqrt{2}-9.8/\sqrt{2}\approx 7.21>0$, rising, as the final part predicts. Also $U/V=\sqrt{2}>\sin\beta=1/\sqrt{5}$.
- Gravity-free intuition: with $g=0$ the particles move in straight lines, and the condition $U/V=\sin\beta/\sin(\theta-\beta)$ is just the sine rule in the triangle formed by A, B and the meeting point. Gravity shifts both particles down by the same amount, so the condition is unchanged.
- A computer check over thousands of random choices of h, d, U, θ confirmed the collision, all three inequalities, and the equivalence in the final part.

Common slips

- Forgetting the ground in part (ii): the inequality comes from the collision happening at a height of at least zero, not from anything in the collision equation.

- Concluding $\theta > \beta$ from $\tan\theta > \tan\beta$ without saying that both angles are acute, where tan is increasing.
- In part (iii), stopping at $\sin(\theta - \beta) \leq 1$; the strict inequality needs $\theta - \beta < \frac{1}{2}\pi$.

Beyond STEP

The cancellation of the $-\frac{1}{2}gt^2$ terms is an instance of changing to a freely falling frame of reference: in a frame accelerating with g, gravity disappears and both particles move in straight lines. This is the simplest case of the Galilean and accelerating-frame ideas that open a first university mechanics course, and later the starting point of Einstein's equivalence principle. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats non-inertial frames carefully.

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